LAB WORKSHEET

ASA Firewall Lab

How a Cisco ASA 5505 decides what to allow: security levels, the default rules, stateful inspection, and using an ACL for exceptions.

Security levels

Every ASA interface has a security level 0–100 — how trusted the network behind it is.

Interface	Level	Meaning
Inside	100	your LAN — most trusted
DMZ	50	public-facing servers
Outside	0	the internet — least trusted

The default rules

- **Higher → lower is allowed** with no rule (Inside→Outside, Inside→DMZ, DMZ→Outside).
- **Lower → higher is denied** by default (Outside→Inside, Outside→DMZ, DMZ→Inside).
- **Stateful inspection:** the ASA tracks connections it allowed out and lets the matching **reply** back in automatically — no rule needed for return traffic.
- **Default-deny inbound:** with no ACL, an unsolicited inbound connection is dropped. "No rule" means deny, not allow.

Making an exception with an ACL

To allow a specific lower→higher flow (e.g. the internet reaching a DMZ web server on 443):

```
access-list OUTSIDE_IN permit tcp any host 203.0.113.10 eq 443
access-group OUTSIDE_IN in interface outside
```

Why a DMZ? A public server sits at level 50, between Outside and Inside. DMZ→Inside is lower→higher, so even a compromised DMZ server cannot start a connection to the trusted Inside LAN.